

Relax Take Home Challenge - Analysis

Goal: Find out which features are associated with user adoption.

Adoption = 3 logins on 3 different days within a 7 day period

Approach: I cleaned and merged user and login data, then created an 'adopted' label. About 13% of users were adopted, so the target was imbalanced. I explored sign-up source, marketing preferences, invitations, and organization. I excluded last_login_time and account lifespan to avoid data leakage.

	accuracy	precision	recall	f1
Logistic Regression	0.44	0.16	0.75	0.26
RF with org_size	0.64	0.15	0.38	0.22
RF without org_size	0.52	0.16	0.62	0.26

Modeling: Logistic Regression was most useful because it had the highest recall, identifying 75% of adopted users. Low precision showed the available features weren't strong predictors. I might explore engineering features for early activity.

Key Findings: Google Authentication and invitations had the strongest positive associations. Personal Project accounts had the strongest negative association. Marketing features were weak predictors. They're associated with 'adoption' but aren't causal.

Limitations/Recommendations: I'd recommend collecting more product-usage and behavior data to improve prediction. Before making any product changes, I'd investigate why Google auth, and invited-users behave differently and test onboarding assistance/tutorials for new users from personal projects to see if that helps increase adoption.

